Quiz

Apr. 13, 2026

多选题 2分

Which of the following statement(s) about page replacement algorithms is/are correct?

A. The OPTIMAL (MIN) algorithm can be implemented in real systems because it predicts future accesses.

B. FIFO can suffer from Bélády’s anomaly – adding more page frames may increase the number of page faults.

C. LRU takes advantage of spatial locality – it replaces the page that has not been accessed many times.

D. RANDOM replacement provides predictable performance and is suitable for hard real-time systems.

E. The Clock algorithm (second-chance) uses a circular list; if the reference bit of the current page is 1, the OS clears it and immediately replaces that page.

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多选题 2分

Which of the following is/are true about LRU approximations (Clock algorithm, Enhanced Clock, Nth-chance)?

A. In the basic Clock algorithm, the reference bit is set by the OS on every memory access.

B. When the Clock algorithm finds a page with reference bit = 1, it clears the bit and moves to the next page.

C. Enhanced Clock uses both a **reference bit** and a **dirty bit**; it selects (0,0) pages first for replacement.

D. The Nth-chance algorithm uses a counter per frame; a larger N makes it a closer approximation to true LRU.

E. The 8-bit shift register method (history counter) exactly implements true LRU without any approximation.

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填空题 2分

In the **8-bit history register** method for approximating LRU, each page frame is associated with an 8-bit counter. Every **100** milliseconds, a timer interrupt occurs. At that time, the **reference bit** is shifted into the [Blank 1] （highest/lowest) bit of the history counter, the other bits shift right, and the [Blank 2] （highest/lowest) bit is discarded. A page that shows a history counter value of **00000000** has not been used for the last [Blank 3] milliseconds.

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主观题 4分

In the context of demand paging, the Clock algorithm (also called second-chance algorithm) is used to approximate LRU page replacement.

1. Briefly describe how the basic Clock algorithm works.

2. Explain how the Enhanced Clock algorithm improves upon the basic Clock algorithm by using a dirty bit.

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多选题 2分

Which of the following statements about kernel and user space memory management in Linux are correct?

A In the standard 32-bit Linux, the kernel virtual addresses range from 0x00000000 to 0xBFFFFFFF, and the user virtual address space starts at 0xC0000000.

B Kernel memory is mapped into the address space of every process so that the OS does not need to switch page tables (change CR3) when handling a system call, interrupt, or exception.

C The kernel memory region is different for each process, which helps isolate one process's kernel data from another's.

D User processes are prevented from accessing kernel space primarily by page table permission bits (e.g., the U/S bit, where U=1 allows user mode access).

E Kernel logical addresses (e.g., from kmalloc()) always map to physically continuous memory starting from 0x00000000 and are never swapped out.

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